# Determinants

<u>Definition.</u> For $A = [a_{ij}] \in \mathrm{Mat}_n(F)$, choose a row $i$. Then

$$\det A = \sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det(A_{ik}),$$

where $A_{ik}$ is the $(n-1)\times(n-1)$ matrix with row $i$, column $k$ removed.

In practice, the $(-1)^{i+k}$ gives

$$\begin{pmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

<u>Theorem.</u> Determinants can also be "expanded along columns" with the sign patterns.

<u>Properties.</u>

If $A$ is a triangular matrix, $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{11} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$

or $A = \begin{pmatrix} a_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$, then $\det A = \prod_{k=1}^{n} a_{kk}$.

<u>Example.</u>

$$\begin{vmatrix} 4 & -1 & 2 & -8 \\ 0 & 3 & 5 & 7 \\ 0 & 0 & 2 & 18 \\ 0 & 0 & 0 & 5 \end{vmatrix} = 4 \begin{vmatrix} 3 & 5 & 7 \\ 0 & 2 & 18 \\ 0 & 0 & 5 \end{vmatrix} = 4 \cdot 3 \begin{vmatrix} 2 & 18 \\ 0 & 5 \end{vmatrix}$$

$$= 4 \cdot 3 \cdot 2 \cdot 5$$

<u>Theorem.</u> For $A, B \in Mat_n(F)$.

i) $\det(AB) = \det A \cdot \det B$

ii) $\det A = \det(A^T)$

iii) $A$ is invertible $\Leftrightarrow$ $\det A \neq 0$.

Proof for iii) ($\Rightarrow$) Suppose $A$ is invertible, then

$$AA^{-1} = I_n \Rightarrow \det(AA^{-1}) = \det(I_n)$$
$$\Rightarrow \det A \det(A^{-1}) = 1$$

Thus, $\det A \neq 0$. Note this also tells us

$$\det(A^{-1}) = \frac{1}{\det A}.$$

($\Leftarrow$) Assume $A$ is not invertible. Let $A \sim M$ be the RREF, i.e. $M = E_k \cdots E_1 A$ for some elementary matrices $E_i$.

$$\Rightarrow A = E_1^{-1} \cdots E_k^{-1} M$$
$$\Rightarrow \det A = \det(E_1^{-1} \cdots E_k^{-1}) \det M.$$

But $A$ is not invertible so $M$ must have a row of zeros.

$$\Rightarrow \det A = \det(E_1^{-1} \cdots E_k^{-1}) \cdot 0$$
$$= 0.$$

## Elementary Row operations and Determinants

**Theorem.**

1. Swapping two rows multiplies the determinant by $-1$.
2. Scaling a row by $\lambda$ multiplies the determinant by $\lambda$.
3. Adding a multiple of a row to another row does not change the determinant.

# Rotations and Reflections Matrices

<u>Definition</u>. Let $\theta \in \mathbb{R}$. Define $R_\theta$ and $T_\theta$ as follows:

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad T_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.$$

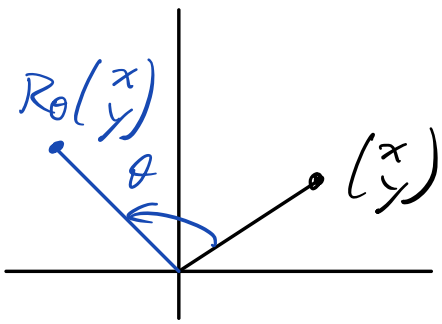
The corresponding matrix transformations are given by

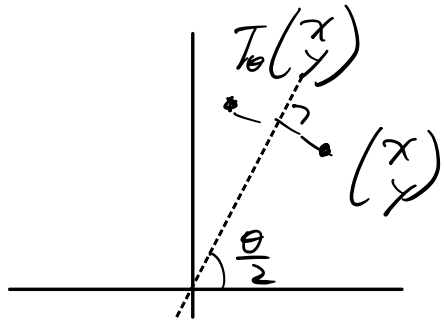
$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto R_\theta \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto T_\theta \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\cos\theta + y\sin\theta \\ x\sin\theta - y\cos\theta \end{pmatrix}$$

- $R_\theta$ rotates the point $(x, y)$ counterclockwise by $\theta$ radians.

- $T_\theta$ reflects across the line making angle $\frac{\theta}{2}$ with positive $x$-axis.




<u>Example</u>. Let $\theta = \frac{\pi}{2}$. Then

$$R_{\frac{\pi}{2}} = \begin{pmatrix} \cos\frac{\pi}{2} & -\sin\frac{\pi}{2} \\ \sin\frac{\pi}{2} & \cos\frac{\pi}{2} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$T_{\frac{\pi}{2}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

<u>Properties</u>.

① $R_0 = R_{2\pi} = I$

② $R_\theta R_\gamma = R_{\theta+\gamma}$

③ $R_\theta^n = R_{n\theta}$

④ $R_\theta^{-1} = R_{-\theta}$

⑤ $T_\theta^2 = I$, so $T_\theta = T_\theta^{-1}$

⑥ $T_\psi^{-1} R_\theta T_\psi = T_\psi R_\theta T_\psi = R_\theta^{-1} = R_{-\theta}$

⑦ $R_\theta T_\psi R_\theta = T_\psi$

⑧ $T_\theta T_\psi = R_{\theta-\psi}$

## Relationships to Dihedral Groups

Let $\theta = \frac{2\pi}{n}$, for some $n \geq 3$. Then $R_\theta$ is a rotation by $\frac{2\pi}{n}$ counterclockwise, $T_\theta$ is a reflection about the line of angle $\frac{\pi}{n}$. Then we get by the general properties

$$R_{\frac{2\pi}{n}}^n = R_{n(\frac{2\pi}{n})} = R_{2\pi} = I, \quad T_{\frac{2\pi}{n}}^2 = I,$$

$$T_{\frac{2\pi}{n}} R_{\frac{2\pi}{n}} T_{\frac{2\pi}{n}} = R_{\frac{2\pi}{n}}^{-1}.$$

**Theorem.** $\langle R_{\frac{2\pi}{n}}, T_{\frac{2\pi}{n}} \rangle \cong D_n$.

**Theorem.** The set $\{R_\theta, T_\theta \mid \theta \in \mathbb{R}\}$ is isomorphic to the symmetry group of a circle.

# Linear Maps

**Definition.** For $n, m \geq 0$, a function $L: F^n \to F^m$ is called linear or a linear map from $F^n$ to $F^m$ if $\forall \vec{v}, \vec{w} \in F^n$ and $\lambda \in F$.

① $L(\vec{v} + \vec{w}) = L(\vec{v}) + L(\vec{w})$,

② $L(\lambda \vec{v}) = \lambda L(\vec{v})$.

## Properties of Linear Maps

① For any linear map $L: F^n \to F^m$, $L(\vec{0}) = \vec{0}$.

② $L: F^n \to F^m$ is linear iff

$$L(\lambda \vec{v} + \mu \vec{w}) = \lambda L(\vec{v}) + \mu L(\vec{w})$$

$\forall \lambda, \mu \in F$, $\forall \vec{v}, \vec{w} \in F^n$.

Proof ①. Let $\lambda = 0 \in F$. $0\vec{v} = \vec{0}$, $\forall \vec{v} \in F^n$.

$\Rightarrow L(\vec{0}) = L(0\vec{v}) = 0L(\vec{v}) = \vec{0}$.

## Composition of Linear Maps

**Proposition.** Let $L_1 : F^n \to F^m$, $L_2 : F^m \to F^p$ be two linear maps. Then

$$L_2 \circ L_1 : F^n \to F^p$$

is also a linear map.

*Proof.* Let $\lambda, \mu \in F$, $\vec{v}, \vec{w} \in F^n$.

$$\begin{aligned}
(L_2 \circ L_1)(\lambda\vec{v} + \mu\vec{w}) &= L_2(L_1(\lambda\vec{v} + \mu\vec{w})) \\
&= L_2(\lambda L_1(\vec{v}) + \mu L_1(\vec{w})) \\
&= \lambda L_2(L_1(\vec{v})) + \mu L_2(L_1(\vec{w})) \\
&= \lambda(L_2 \circ L_1)(\vec{v}) + \mu(L_2 \circ L_1)(\vec{w})
\end{aligned}$$

**Definition (Identity Map).**

The identity map $i : F^n \to F^n$ is given by $i(\vec{v}) = \vec{v}$.

**Definition (Invertible Map).** We say a linear map $L : F^n \to F^n$ is invertible if there

is some linear map $T : F^n \to F^n$ such that

$$L \circ T = T \circ L = i : F^n \to F^n.$$

Then we write $T = L^{-1}$.

<u>Example</u>. $L : \mathbb{Z}_7^2 \to \mathbb{Z}_7^2$,

$$L(x,y) = (3x+4y,\ x+2y)$$

Then $L$ is invertible with

$$L^{-1}(x,y) = (x+5y,\ 3x+5y)$$